

Physics-Constrained, Source-Aware Spatio-Temporal Graph Neural Networks for Atmospheric Transport: A Methodological Investigation of 15 Foundational Studies

Foundational Paradigm Shifts in Atmospheric Machine Learning

Atmospheric particulate matter forecasting across densely populated, transboundary basins—exemplified by the Indo-Gangetic Plain and the National Capital Region (NCR) of Delhi—presents fundamental challenges to conventional statistical learning¹. Historically, predictive systems have occupied two distinct, incompatible paradigms: deterministic chemical transport models (CTMs) and unconstrained deep spatio-temporal graph neural networks (STGNNs)¹. Deterministic Eulerian and Lagrangian CTMs (such as WRF-Chem and CMAQ) explicitly integrate the partial differential equations governing fluid advection, turbulent diffusion, dry and wet deposition, and non-linear chemical transformations⁵. However, these numerical systems suffer from severe computational overhead, rigid grid discretization, and high sensitivity to uncertain, lagging emission inventories¹.

Conversely, deep learning models represent monitoring stations as nodes in a graph, attempting to learn temporal auto-correlations and spatial dependencies through unconstrained message passing and attention architectures¹. While computationally efficient, standard STGNNs fail during extreme, non-stationary pollution episodes¹. By relying on static Euclidean distance matrices or unconstrained attention weights, these networks treat the atmosphere as an isotropic, closed correlation system¹. They fail to capture the physical reality of atmospheric transport: fluid flow is highly directional, turbulent mixing is non-linearly governed by boundary layer dynamics, transboundary plumes require physical travel time to traverse geographic space, and total pollutant mass must obey conservation principles¹.

To resolve these architectural limitations, a new scientific paradigm has emerged: physics-constrained, source-aware neural transport networks¹. Rather than delegating spatial reasoning entirely to unconstrained data-driven layers, this framework embeds the partial differential equations of fluid mechanics and atmospheric chemistry directly into the graph message-passing operators, neural ordinary differential equations, and loss regularizers⁴. Fifteen foundational research publications provide the methodological, mathematical, and structural building blocks for this operational architecture¹. These studies address five core challenges: the numerical discretization of coupled advection-diffusion-reaction operators;

open-system boundary dynamics and continuous-time neural state spaces; dynamic topological reconfiguration governed by atmospheric stability; multi-scale spatial hierarchies that resolve transport travel delays; and explainable source attribution heads coupled with event-aware regime-switching mechanisms¹.

Advection-Diffusion-Reaction Formulations and Differential Equation Networks

Coupled Advection-Diffusion-Reaction Operators in Graph Convolutions

The fundamental physical process governing pollutant dispersal across space and time is the atmospheric advection-diffusion-reaction (ADR) partial differential equation¹. In TransNet, Dimri et al. (2026) address the limitations of empirical graph attention by directly embedding coupled ADR numerical operators into the convolutional layers of a graph neural network designed for multi-station particulate matter forecasting⁴. The network models the scalar concentration

field $C(\mathbf{x}, t)$ according to the continuous transport balance:

$$\frac{\partial C}{\partial t} = -\nabla \cdot (\mathbf{u}C) + \nabla \cdot (K\nabla C) + R(C)$$

where $\mathbf{u} = (u, v)$ denotes the horizontal wind velocity field, K represents the turbulent eddy diffusivity tensor, and $R(C)$ accounts for sink mechanisms and chemical transformations¹.

Rather than executing generic feature transformations, TransNet establishes a directed computational graph across 170 monitoring stations where edge weights reflect directional projections of instantaneous wind vectors onto inter-station lines of sight⁴. TransNet decouples message passing into two distinct structural operators: an asymmetric advection operator that transports latent mass strictly downwind, and a symmetric Laplacian diffusion operator that dissipates mass across regional spatial gradients⁴.

Crucially, TransNet introduces an operator ablation analysis demonstrating the structural necessity of physical coupling⁴. When the network is trained with isolated advective or isolated diffusive branches, prediction error increases by 72% to 73%⁴. Completely removing the coupled transport operator leads to a 153.5% error increase at early prediction lead times⁴. This framework demonstrates that integrating fluid dynamics directly into message-passing layers provides stronger inductive biases than classical attention mechanisms, creating a proven blueprint for modeling directional pollutant plumes¹.

Differential Equation Networks for Decoupled Advective-Diffusive Message Passing

While TransNet structures message passing around discrete numerical approximations of transport, AirPhyNet (Hettige et al., 2024) formulates atmospheric particle movement through continuous differential equation networks integrated into graph architectures¹². AirPhyNet resolves the vulnerability of purely data-driven models to sparse, irregularly distributed monitoring stations and abrupt meteorological transitions¹².

The authors map monitoring sites to graph vertices and construct a dual-edge topological configuration: static geographic distance edges that constrain isotropic background diffusion, and dynamic meteorological edges that represent wind-driven advection¹². AirPhyNet explicitly decouples the governing differential equations into parallel latent branches, allowing independent parameterization of directional advective flux and spatial gradient diffusion¹². By evaluating these differential networks over multi-step integration intervals, the model preserves physical consistency across 24-hour, 48-hour, and 72-hour forecasting horizons¹². Evaluated on multi-city operational benchmarks, AirPhyNet demonstrated a 10% reduction in prediction error relative to unconstrained STGNN baselines¹². Its primary methodological contribution lies in its robustness under data sparsity: when intermediate monitoring stations experience sensor dropouts, the underlying differential equation constraints prevent the network from generating spurious spatial artifacts, ensuring physically plausible mass trajectories across unmonitored intervening terrain¹².

Continuous-Field Chemical Transport in Eulerian Neural Networks

A structural limitation common to both TransNet and AirPhyNet is their restriction to graph nodes anchored at discrete monitoring stations⁴. In CTENet, Zhang et al. (NeurIPS 2025) address this limitation by moving from Lagrangian station graphs to a continuous Eulerian neural framework informed by chemical transport principles¹³.

CTENet conceptualizes air pollution not as an abstract set of discrete time series, but as an evolving two-dimensional continuous spatial field¹⁴. The framework comprises three coordinated components: an Eulerian Pollution Encoder that maps sparse, point-based observations into a continuous coordinate space; a Meteorological Encoder that extracts high-dimensional representations of continuous wind velocity, temperature, and planetary boundary layer fields; and an Eulerian ADR Decoder that solves the governing transport equations over continuous spatio-temporal coordinates¹⁴. The decoder explicitly incorporates the two-dimensional atmospheric advection-diffusion-reaction equation:

$$\frac{\partial C}{\partial t} = - \left(u \frac{\partial C}{\partial x} + v \frac{\partial C}{\partial y} \right) + \frac{\partial}{\partial x} \left(K_x \frac{\partial C}{\partial x} \right) + \frac{\partial}{\partial y} \left(K_y \frac{\partial C}{\partial y} \right) + S_C(\mathbf{x}, t)$$

where $S_C(\mathbf{x}, t)$ incorporates source injection and atmospheric sink terms¹⁴.

By embedding physics-informed neural network (PINN) regularization across spatial collocation points, CTENet penalizes mass-flux inconsistencies at regional boundaries¹⁴. Across

continental-scale networks in the United States and China, CTENet achieved an RMSE improvement of 45.8% on the US dataset and 21.0% on the Chinese network compared to state-of-the-art baselines¹³.

The continuous formulation of CTENet provides the theoretical foundation for incorporating unmonitored external source zones—such as rural agricultural fires—into urban monitoring frameworks without requiring artificial station interpolation¹.

Tracer-Transport Mass Regularization and Spatial Generalization

Building on continuous transport principles, Assiotis et al. (PMLR 2025) examine how urban atmospheric tracer transport equations can serve as inductive regularizers in graph convolutional networks paired with gated recurrent units (PI-GCN-GRU)¹⁸. Focus is placed on urban nitrogen dioxide and reactive particulates, treating the airborne mass as a dynamic tracer field governed by localized wind vectors and surface roughness lengths¹⁸.

The spatial message passing explicitly incorporates an advection-diffusion conservation constraint directly into the loss function¹⁸:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{forecast}} + \alpha \sum_i \left| \frac{\Delta C_i}{\Delta t} + \nabla \cdot \mathbf{J}_i \right|$$

where $\mathbf{J}_i$ denotes the total physical flux entering or exiting the urban control volume centered at station i ¹⁸.

Assiotis et al. evaluate their framework under strict spatial-holdout conditions, where entire monitoring clusters are withheld during optimization¹⁸. While purely data-driven STGNNs suffered catastrophic performance degradation when evaluating unseen geographic coordinates, the physics-informed model retained its predictive skill¹⁸.

This demonstrates that embedding transport-mass conservation constraints into graph aggregation enables zero-shot spatial generalization, providing a vital tool for expanding predictions into unmonitored peri-urban and rural areas¹.

Open-System Boundary Dynamics, Continuous Latent Spaces, and Generative Regularization

Boundary-Aware Dual Neural Ordinary Differential Equations

A persistent flaw in applying classical physics-informed machine learning to urban airsheds is the assumption of a closed system¹⁰. Standard PINN architectures implicitly assume that total mass is conserved within a static bounding box³⁰. In reality, urban airsheds are open thermodynamic systems subject to unmonitored external inflow, downwind outflow, stochastic ground emissions, and complex boundary layer entrainment¹⁰.

In Air-DualODE, Tian et al. (ICLR 2025) resolve this mismatch through a dual-branch Neural

Ordinary Differential Equation architecture¹⁰. The primary branch solves a discrete Boundary-Aware Diffusion-Advection Equation (BA-DAE), designed specifically to model open-boundary conditions¹⁰:

$$\frac{dC_i(t)}{dt} = \mathcal{F}_{\text{adv}}(C_i, \mathbf{u}_i) + \mathcal{F}_{\text{diff}}(C_i, K) + S_i(t) - D_i(t) + \Phi_{\text{boundary},i}(t)$$

where $\Phi_{\text{boundary},i}(t)$ calculates explicit mass flux across the airshed boundary, while $S_i(t)$ and $D_i(t)$ govern station-specific emission generation and surface dissipation¹⁰.

Operating in parallel, a high-capacity data-driven Neural ODE branch captures unmodeled residual dynamics, including complex secondary photochemical reactions and stochastic micro-scale traffic turbulence¹⁰. The latent states of both branches are dynamically projected into a shared metric space and unified through an adaptive temporal fusion gate¹⁰.

Air-DualODE establishes that open-system boundary modeling is mandatory for regional forecasting: isolating boundary influx allows the architecture to forecast sudden pollution spikes driven by transboundary advection without violating physical mass balances¹⁰.

Multi-Timescale Latent State Partitioning and Uncertainty Calibration

Atmospheric pollutant trajectories exhibit complex multi-scale temporal dynamics: localized mobile source emissions fluctuate on sub-hourly scales, planetary boundary layer heights oscillate diurnally, and synoptic weather systems evolve over multi-day synoptic periods³³. Discrete-time recurrence models (such as LSTMs and GRUs) accumulate substantial numerical integration errors across long multi-step rollouts and cannot naturally process irregular sampling intervals or sensor dropouts³³.

In MA-NODE, Havaei, Shahhosseini, and Maknoon (2025) address these temporal limitations by implementing continuous-time Neural ODEs featuring multi-timescale latent space

decomposition³⁴. The complete latent representation $\mathbf{z}(t) \in \mathbb{R}^{128}$ is partitioned into three decoupled subspaces:

$$\mathbf{z}(t) = [\mathbf{z}_{\text{fast}}(t), \mathbf{z}_{\text{medium}}(t), \mathbf{z}_{\text{slow}}(t)]$$

Each subspace evolves according to independent parameterized neural vector fields:

$$\frac{d\mathbf{z}_k(t)}{dt} = f_{\theta_k}(\mathbf{z}_k(t), \mathbf{x}(t)), \quad k \in \{\text{fast}, \text{medium}, \text{slow}\}$$

where distinct neural networks model sub-hourly traffic peaks, diurnal boundary layer breathing, and multi-day synoptic weather fronts, respectively³³. An integrated multi-head attention mechanism continuously evaluates queries and keys across these subspaces, adaptively emphasizing rapid dynamic responses during pollution episodes while preserving slow baseline trends during stable periods³³.

Furthermore, MA-NODE integrates continuous-time negative log-likelihood loss functions to yield well-calibrated 95% predictive uncertainty intervals (achieving an empirical coverage of approximately 0.90) alongside point forecasts with $R^2 >$ across multi-step horizons³⁴. This framework demonstrates how continuous-time ordinary differential equations can model multi-scale temporal dynamics while providing operational uncertainty quantification¹.

Partial Differential Equation Regularization in Denoising Diffusion Models

Generative diffusion models possess high capacity for characterizing complex, multi-modal probability distributions of high-dimensional spatio-temporal fields²³. However, unconstrained reverse sampling often produces physically implausible spatial gradients and unrealistic high-frequency localized noise²³.

In STeP-Diff, Tian et al. (IEEE TKDE 2025/2026) couple conditional Denoising Diffusion Probabilistic Models (DDPMs) with Deep Operator Networks (DeepONet) to generate continuous urban pollution fields under sparse, mobile sensor coverage²³. To ensure physical realism, STeP-Diff introduces a PDE-constrained regularization framework directly into the generative reverse denoising process²³.

The optimization objective couples standard conditional variational lower-bound losses with an explicit convection-diffusion penalty:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{diff}}(\theta) + \gamma \left\| \frac{\partial \hat{C}}{\partial t} + \mathbf{u} \cdot \nabla \hat{C} - K \nabla^2 \hat{C} - S \right\|_2^2$$

Through this formulation, the denoising trajectory asymptotically converges toward states that satisfy fundamental physical laws²³.

STeP-Diff confirms that PDE-based physical constraints can regularize expressive generative models, preventing hallucinations and preserving spatial continuity even when up to 30%–50% of monitoring inputs are randomly missing²³.

Dynamic Topological Reconfiguration Governed by Micrometeorology and Stability

Dynamic Geographical Graph Construction via Wind Vector Projections

Graph neural networks frequently fail to outperform standard recurrent baselines when their spatial topologies are fixed to static geographic distance metrics¹. Zhao et al. (2025) address this issue by proposing a dynamic geographical Graph Neural Network (GNN) that constructs time-varying directional graphs directly from real-time wind-field vectors².

The model pairs a static geographic graph $\mathbf{A}_{\text{static}}$, which captures permanent topographical proximity, with a dynamic directional adjacency matrix $\mathbf{A}_{\text{dynamic}}(t)$ ². The dynamic directional edge between station i and station j at time step t is calculated via:

$$A_{ij}^{\text{dynamic}}(t) = \max \left(0, \frac{\mathbf{u}_i(t) \cdot (\mathbf{x}_j - \mathbf{x}_i)}{\|\mathbf{u}_i(t)\| \|\mathbf{x}_j - \mathbf{x}_i\|} \right) \cdot \exp \left(-\frac{\|\mathbf{x}_j - \mathbf{x}_i\|^2}{2\sigma_d^2} \right) \cdot \phi(\|\mathbf{u}_i(t)\|)$$

where $\mathbf{u}_i(t)$ represents the 10-meter wind vector at the source node, $\mathbf{x}$ denotes coordinate locations, and $\phi(\|\mathbf{u}_i(t)\|)$ scales the edge weight based on wind velocity². When the wind blows from station i toward station j , an edge $i \rightarrow j$ is activated; when the wind shifts in the opposite direction, the edge weight drops to zero².

Zhao et al. combine this dynamic directed graph with a Graph Convolutional Network and Long Short-Term Memory (GCN-LSTM) architecture². Across regional air quality networks in China and India, the dynamic directional graph demonstrated clear performance advantages over static spatial baselines, confirming that time-varying wind fields must govern spatial message passing².

However, this formulation models transport purely as a directional geometric alignment; it neglects the critical regulatory role of boundary layer height and vertical thermal stability on horizontal edge propagation¹.

Structural Inductive Biases via Heterogeneous Dynamic Operator Decoupling

Taking dynamic topology learning further, UrbanGraph (Xin et al., ICLR 2026 / arXiv 2025) introduces structural inductive biases by transforming physical first principles directly into time-varying causal graph topologies¹⁵. UrbanGraph argues that allowing deep models to learn spatial relationships implicitly through latent attention leads to physical inconsistencies and excessive computational overhead¹⁵.

The framework constructs dynamic heterogeneous graphs where edge configurations update each hour based on underlying physical phenomena, including anisotropic wind convection and solar radiation gradients¹⁵. UrbanGraph introduces a specialized message-passing mechanism that decouples physical operators:

$$\mathbf{h}_i^{(l+1)} = \sum_{\tau \in \mathcal{T}} \sum_{j \in \mathcal{N}_i^\tau(t)} \mathcal{W}_\tau \left(\mathbf{h}_j^{(l)}, \mathbf{e}_{ij}^\tau(t) \right)$$

where $\mathcal{T} = \{\text{convective, diffusive, thermal}\}$ represents distinct physical interaction channels¹⁵. By enforcing dynamic topological changes as hard structural constraints, the model aligns its receptive field strictly with the time-varying physical domain of influence¹⁵.

Compared to conventional spatio-temporal baselines, UrbanGraph improved predictive accuracy by up to 10.8% in R^2 while simultaneously reducing computational complexity by 17.0% in FLOPs¹⁵. Controlled ablations revealed that dynamic topological reconfiguration contributed a 7.1% accuracy gain, while operator decoupling provided an additional 3.5% gain³⁶. This demonstrates that enforcing dynamic physical topologies provides both higher predictive accuracy and improved computational efficiency¹⁵.

Multi-Faceted Atmospheric Diffusion Edges for Extended Forecasting Horizons

In long-term particulate forecasting, the effective spatial correlation radius changes dynamically with lead time¹⁶. Kim, Jin, and Suk (2023) examine this phenomenon, noting that static distance metrics trigger severe performance degradation beyond 24-hour lead times due to over-smoothing and localized dependency bottlenecks¹⁶.

The authors develop a multi-gated spatio-temporal graph neural network utilizing multi-faceted edge formulations for mid-to-long-term forecasting up to 96 hours¹⁶. Edges are synthesized by combining an empirical concentration similarity metric with an analytical atmospheric diffusion coefficient¹⁶:

$$A_{ij}(t, \Delta t) = \alpha \cdot \exp \left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2K_{\text{diff}}(\mathbf{u}_t)\Delta t} \right) + (1 - \alpha) \cdot \text{Sim}(C_i, C_j)$$

where K_{diff} is derived from the localized wind velocity field and atmospheric turbulence, and Δt corresponds to the forecasting lead time¹⁶. As the forecast horizon extends, the spatial receptive field of the graph expands dynamically to match the broader physical domain of diffusion¹⁶.

Evaluating 96-hour particulate forecasts across South Korea, this multi-faceted edge formulation improved the Critical Success Index (CSI) by 3.14% and increased R^2 by 5.26%

over standard temporal baselines¹⁶.

This confirms that the effective edge length within atmospheric transport graphs cannot remain static, but must scale proportionally with transport velocity and forecast lead time¹.

Multi-Scale Spatial Hierarchies and Source-to-Receptor Delay Dynamics

Coupling AI Global Weather Predictions with Hierarchical Graph Neural Networks

To extend reliable forecasts into multi-day horizons, local station observations must be coupled with synoptic-scale numerical meteorological forecasts¹⁹. In a recent framework published in ACS / PMC12951247 (2025/2026), researchers couple Google DeepMind's GraphCast—an

AI-based global weather model producing 0.25° resolution forecasts from ERA5

reanalysis—with a hierarchical graph neural network for medium-range $\text{PM}_{2.5}$ forecasting¹⁹.

The architecture addresses the spatial scale mismatch between synoptic weather systems and localized urban pollution¹⁹. Gridded meteorological fields (temperature, geopotential height,

and u/v wind components at multiple vertical pressure levels) are processed through a ResNet-50 spatial backbone paired with an LSTM temporal encoder¹⁹. Simultaneously, surface air monitoring stations are clustered based on mutual statistical correlation and geographical proximity to establish a two-level hierarchical graph¹⁹.

Within each cluster, a local Graph Attention Network models micro-scale intra-cluster diffusion¹⁹. Local representations are then pooled into virtual regional super-nodes that form a global inter-cluster graph capturing large-scale transboundary transport¹⁹. Fusing the local and global spatial representations through cross-attention mechanisms enabled accurate autoregressive multi-day particulate forecasting¹⁹.

This validates a hierarchical graph design: rather than propagating messages across a flat, fully connected network of stations, models should separate intra-urban diffusion from regional corridor transport¹.

Dynamic Source-to-Receptor Attribution and Wind-Driven Transport Lags

A widespread design flaw in urban graph neural networks is the implicit assumption of instantaneous spatial influence: when an edge is activated between an upwind node and a downwind node, standard graph convolutions propagate information within a single time step¹.

In reality, airborne mass requires hours to traverse regional distances¹.

Hu et al. (2023) examine this delay dynamic using a hybrid GNN-LSTM framework designed for 72-hour particulate forecasting across Beijing and 128 surrounding regional monitoring stations¹⁷. The authors track the temporal evolution of learned spatial attention weights between distant upwind emission sites and downwind urban receptor nodes across progressive forecasting horizons¹⁷.

The experimental data revealed an important empirical finding: at immediate short-term horizons (+1 to +3 hours), the influence of distant regional stations on urban receptor concentrations was approximately 0%¹⁷. However, as the forecast lead time extended along prevailing downwind pathways, the contribution of regional transport grew steadily, explaining 38% of total urban particulate concentration variance at the 72-hour horizon¹⁷. This empirical finding demonstrates that source-to-receptor influence in atmospheric systems is governed by continuous, time-lagged transport dynamics¹. It provides direct justification for implementing an explicit dynamic transport lag formulation:

$$\tau_{s,i}(t) = \frac{d_{s,i}}{U_{\text{eff}}(t)}$$

where $d_{s,i}$ is the transport distance between source s and receptor i , and $U_{\text{eff}}(t)$ is the effective path-integrated wind transport velocity¹.

Multi-Scale Multi-Graph Spatial Correlation Networks

To capture the distinct physical mechanisms operating at different spatial resolutions, Bo et al. (2024) developed MGAtt-LSTM, a multi-scale spatial correlation prediction model based on multi-graph attention networks²¹. The authors show that attempting to encode geographic distance, weather similarities, and regional topology into a single adjacency matrix leads to conflicting gradient updates and poor model convergence²¹.

MGAtt-LSTM constructs three parallel, decoupled spatial graphs: a geographic distance graph that models localized proximity; a meteorological similarity graph that links stations sharing comparable temperature, humidity, and barometric conditions; and a regional connectivity graph that reflects synoptic transport corridors²¹. The architecture deploys independent multi-head graph attention mechanisms across each matrix:

$$\mathbf{z}_{\text{multi}} = \bigoplus_{m \in \{\text{geo}, \text{met}, \text{reg}\}} \sigma \left(\sum_{j \in \mathcal{N}_i^m} \alpha_{ij}^m \mathbf{W}_m \mathbf{h}_j \right)$$

where $\bigoplus$ denotes tensor concatenation followed by a linear projection gate²¹.

Evaluated across complex monitoring basins, MGAtt-LSTM achieved significant reductions in MAE and RMSE compared to single-graph GNN baselines²¹. The multi-graph architecture proved particularly resilient during winter stagnation events, when regional synoptic transport stalled while localized geographic accumulation surged²¹.

This confirms that atmospheric models require multi-layered graph representations to

decouple localized accumulation from regional transport¹.

Real-Time Source Apportionment, Non-Linear Chemistry, and Episodic Regimes

Deep Response Surface Surrogate Modeling for Real-Time Source Attribution

Air quality managers require more than single-point numerical predictions; effective intervention requires knowing *why* concentrations are elevated and quantifying sector-specific emission contributions¹. Traditional source apportionment relies on receptor models (e.g., Positive Matrix Factorization) or computationally expensive chemical transport sensitivity simulations (e.g., brute-force emission zero-out runs)¹.

In DeepRSM, Xing et al. (2020) introduce a deep learning response surface surrogate architecture that predicts non-linear air quality responses to sector- and region-specific emission changes in real time²⁴. DeepRSM serves as a high-speed neural emulator for three-dimensional chemical transport models (WRF-CMAQ), taking localized emission perturbation factors across multiple economic sectors alongside meteorological boundary inputs to predict future concentrations²⁴:

$$C_{\text{target}}(\mathbf{x}) = f_{\theta}(\Delta E_{\text{traffic}}, \Delta E_{\text{industry}}, \Delta E_{\text{power}}, \Delta E_{\text{agriculture}}, \mathbf{M}_{\text{met}})$$

By computing partial derivatives of the learned response function with respect to individual sectoral emission inputs, DeepRSM derives real-time marginal source contributions:

$$\text{Contribution}_k = \frac{\partial C_{\text{target}}}{\partial E_k} \cdot E_k$$

Crucially, DeepRSM captures the non-linear chemical regimes that govern secondary inorganic and organic aerosol formation, accurately modeling shifts between VOC-limited and NO_x -saturated chemical states²⁴. Validated across severe regional haze episodes in northern China,

DeepRSM reproduced 3D chemical transport model responses with $R^2 >$ while running orders of magnitude faster²⁴.

This framework provides the methodological foundation for integrating a real-time "Source Contribution Head" directly into neural forecasting architectures, enabling simultaneous concentration forecasting and quantitative source attribution¹.

Deep Transfer Learning and Attention Dynamics During Winter Stubble Burning

In the Indo-Gangetic Plain, seasonal post-monsoon and winter air pollution peaks are heavily driven by regional agricultural residue burning in Punjab and Haryana, which interacts with urban emissions under low boundary layer heights¹. In Lakshmi and Krishnamoorthy (2025), the authors develop a transfer learning and multi-head attention forecasting system designed specifically for the Delhi stubble-burning window (September through December) using a ten-year dataset (2012–2022)⁹.

The architecture integrates lagged ground-level pollutant measurements, high-resolution meteorological variables (wind speed, wind direction, planetary boundary layer dynamics), and regional satellite fire detections⁹. To overcome severe data distribution shifts between normal seasons and episodic peak periods, the framework utilizes a two-phase transfer learning pipeline: pre-training across multi-year background observations to learn baseline atmospheric chemistry, followed by fine-tuning on extreme winter pollution episodes⁹.

The authors conducted attention-head visualizations that revealed distinct operational modes across individual heads⁹. Specific attention heads remained inactive during calm meteorological conditions but activated strongly when elevated upwind fire activity coincided with northwesterly wind vectors directed toward Delhi, directly capturing transboundary smoke advection⁹.

This study confirms the necessity of engineering specialized fire activity metrics (such as Fire Radiative Power proxies) and confirms that models must be tailored to capture the episodic dynamics of the stubble-burning season¹.

Dynamic Edge Learning and Event-Aware Architectures

Dynamic Adjacency Generation via Meteorological Encodings

During seasonal transitions, atmospheric circulation patterns frequently shift from high-velocity regional ventilation to total stagnation²². Static graph structures cannot capture these abrupt changes in physical connectivity⁸. Dun, Yang, and Lei (2022) address this challenge by developing a Dynamic Graph Convolutional Network coupled with Multi-Channel Temporal Convolutional Networks (DGC-MTCN) for urban air quality forecasting³⁸.

DGC-MTCN computes dynamic adjacency weights at each hourly time step as an endogenous function of evolving node states:

$$\mathbf{A}_{\text{dyn}}(t) = \text{Softmax} \left(\text{ReLU} \left(\mathbf{E}(t) \mathbf{W}_e \mathbf{E}(t)^T \right) \right)$$

where $\mathbf{E}(t) \in \mathbb{R}^{N \times d}$ encodes real-time wind speed, wind direction, localized barometric pressure, and temperature gradients³⁸. When atmospheric conditions are well-mixed and windy, the dynamic edge generation function activates long-range inter-station connections³⁸. Conversely, when localized stagnation occurs, long-range edges are pruned, collapsing the graph into localized self-loops³⁸.

By coupling dynamic spatial message passing with multi-channel temporal convolutions, DGC-MTCN outperformed static graph baselines, demonstrating superior tracking of sudden pollution build-ups during stagnant weather³⁸.

This framework provides an architectural mechanism for dynamically updating graph connectivity based on real-time atmospheric states¹.

Frequency-Aware Period Folding and Decoupled Industrial Graph Backbones

Urban air pollutants exhibit fundamentally different temporal dynamics: photochemical oxidants like ground-level ozone display strong diurnal solar cycles, primary combustion pollutants like carbon monoxide follow localized traffic rush hours, and fine particulate matter combines continuous background accumulation with sudden episodic spikes²⁰. In FFTGNet (CMES 2025/2026), researchers propose a frequency-aware spatio-temporal graph neural network to address these cross-scale temporal dynamics in multi-pollutant systems²⁰. FFTGNet incorporates a Fast Fourier Transform (FFT) dominant-period estimation module that transforms incoming time series into the spectral domain, extracts primary periodic components, and executes period-folding prior to spatial message passing²⁰. The spatial-temporal backbone then pairs Chebyshev graph convolutions with Temporal Gated Linear Units (TemporalGLU)²⁰. Decoupling recurring diurnal variations from transient episodic spikes in the frequency domain prevents extreme pollution events from destabilizing the learned baseline representations²⁰.

Across multi-step forecasting horizons (1 to 6 hours), FFTGNet achieved the highest R^2 and lowest RMSE across multiple pollutants, outperforming competitive baseline models like MGCGRU-SAN and AirNN²⁰.

The period-folding and frequency-decomposition strategy of FFTGNet offers a principled approach to decouple predictable diurnal boundary layer oscillations from sudden, non-periodic pollution episodes like crop burning or dust storms¹.

Multi-Source Characterization of Stagnation and Thermal Inversion in Delhi-NCR

Air quality modeling within the Delhi-NCR airshed requires addressing severe meteorological stagnation, where winter thermal inversions trap pollutants near the surface¹. In RegBoost, the Gurugram Air Research Initiative (2024/2025) investigates particulate dynamics across Gurugram and the broader Delhi-NCR region from 2019 to 2023². The study integrates ground-level CAAQMS monitoring networks, ERA5 meteorological and planetary boundary layer height reanalysis, and satellite observations from Sentinel-5P TROPOMI and NASA MODIS/VIIRS¹⁵.

The authors present detailed empirical analyses of winter thermal inversion episodes²². They show that when low ambient surface temperatures, elevated relative humidity, and calm winds coincide with boundary layer heights falling below 150 meters, vertical atmospheric mixing ceases, leading to extreme localized pollutant accumulation²². RegBoost utilizes an iterative spatial imputation framework coupled with gradient boosting to handle sensor outages and

model non-linear pollutant accumulation during stagnation events²². This empirical study demonstrates that modeling urban air pollution in Delhi-NCR requires incorporating continuous boundary layer height and stability indicators to regulate spatial dispersion¹.

Comparative Methodological Analysis of the 15 Foundational Studies

The following structured matrix compares the mathematical foundations, spatial representations, physical drivers, and validation strategies across the 15 foundational research publications:

Investigation & Primary Citation	Methodological Domain & Architectural Backbone	Governing Mathematical Formulation / Constraint Mechanism	Spatial & Topological Graph Representation	Atmospheric, Meteorological & Satellite Drivers	Empirical Benchmarks & Validation Design
TransNet (Dimri et al., 2026, npj Clean Air) [cite: 1, 4]	Transport-Informed Graph Neural Network	$\frac{\partial C}{\partial t} = -\nabla \cdot$ embedded into graph convolutions	Directed dynamic graph based on wind vector alignments	Horizontal wind fields $\mathbf{u} =$, turbulent eddy diffusivity K	170 operational stations; +72-h rollouts; operator ablation (153.5% error jump without coupled ADR) ⁴
AirPhyNet (Hettige et al., 2024, ICLR) [cite: 12, 43]	Physics-Guided Differential Equation GNN	Coupled differential equation networks parameterizing advection and	Static distance topology combined with dynamic wind-velocity graph	Wind speed, direction, inter-station spatial temperature gradients	Multi-city benchmarks; missing data stress-tests; 24–72 h lead times (10% error

		diffusion equations			drop) ¹²
CTENet (Zhang et al., 2025, NeurIPS) [cite: 13, 14]	Continuous Eulerian Chemical Transport PINN	2D Eulerian chemical transport ADR equation via continuous neural decoders	Continuous 2D Eulerian field (coordinate continuous space)	Continuous wind vector fields, satellite thermal inversion profiles, boundary fluxes	Continental-scale US and China networks; RMSE drops of 45.8% (US) and 21.0% (China) ¹³
PI-GNN (Assiotis et al., 2025, PMLR) [cite: 18, 28]	Physics-Informed Urban Tracer GCN-GRU	Mass-conservation tracer regularizer: $\mathcal{L}_{\text{total}} = \mathcal{L}_M$	Dynamic directed graph scaled by physical transport velocity	10-m horizontal winds, surface friction velocity, urban roughness lengths	Nationwide Netherlands network; spatial holdout validation on unobserved stations ¹⁸
Air-DualODE (Tian et al., 2025, ICLR) [cite: 10, 30]	Open-System Dual-Branch Neural ODE	Discrete Boundary-Aware Diffusion-Advection Equation (BA-DAE) + Data Neural ODE	Open-system boundary topology with explicit boundary flux terms	Ambient wind fields, boundary source influx, localized surface deposition rates	Urban monitoring networks; boundary flux perturbation ablation experiments ¹⁰
MA-NODE (Havaei et al., 2025, Sci. Rep.) [cite: 34, 35]	Multi-Timescale Continuous Latent Neural ODE	Tri-timescale latent ODE partitioning: $[\mathbf{z}_{\text{fast}}, \mathbf{z}_{\text{med}}, \mathbf{z}_{\text{slow}}]$ with attention	Continuous-time station-level latent trajectories	Planetary Boundary Layer Height (BLH), humidity, temperature, surface	Multi-step rollouts; $R^2 > 0.9$; calibrated 95% uncertainty interval

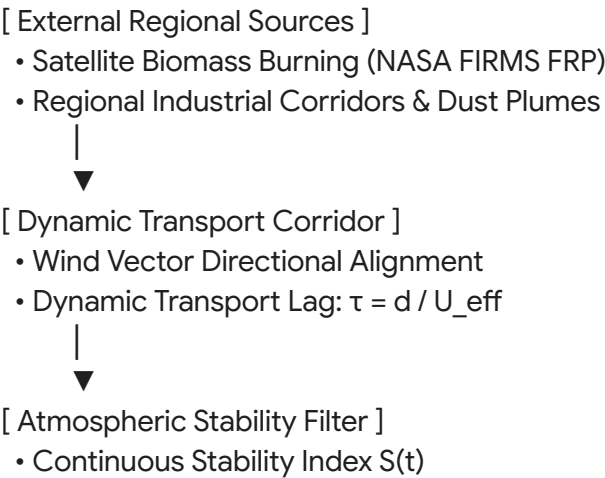
		synthesis		pressure	coverage ($\approx$) ³⁴
STeP-Diff (Tian et al., 2026, IEEE TKDE) [cite: 23]	PDE-Regularized Generative Diffusion Model	DeepONet operator mapping + DDPM denoising penalized by convection-diffusion PDE	Continuous coordinate subregions via mobile and stationary nodes	Localized advective wind vectors, turbulent kinetic energy fields	Mobile sensor deployment s; stable reconstruction under 30%–50% missing data ²³
Zhao et al. (2025, Env. Modell. Softw.) [cite: 2]	Dynamic Geographical GCN-LSTM	Dynamic directional wind-projection matrix $\mathbf{A}_{\text{dyn}}(t)$ combined with static distance graph	Dual graph: Static geographic distance + Dynamic wind directional alignment	Hourly 10-meter wind vectors, inter-station Euclidean distance	Regional monitoring networks in China and India; lead-lag dispersion tests ²
UrbanGraph (Xin et al., 2026, ICLR) [cite: 15, 36]	Dynamic Heterogeneous Operator-Decoupled GNN	Physical operator decoupling: $\mathbf{h}_i^{(l+1)} =$ over advection, diffusion, thermal	Heterogeneous dynamic graph with time-evolving topologies	Anisotropic wind convection, solar radiation shading, urban morphology	Urban microclimate networks; +10.8% R^2 gain; 17.0% FLOP reduction ¹⁵
Kim et al. (2023, J. Clean. Prod.) [cite: 16]	Multi-Gated Spatio-Temporal Graph Network	Multi-faceted edge formulation combining diffusion coefficients	Multi-edge dynamic graph scaling with forecasting lead time	Atmospheric turbulence diffusion coefficients, wind	96-hour extended horizons; Critical Success Index (CSI)

		with feature similarity		velocity, distance	+3.14%; R^2 +5.26% ¹⁶
GraphCast-GNN (2025/2026, ACS) [cite: 19]	Weather-Coupled Hierarchical STGNN	ResNet-LSTM weather feature extraction coupled with multi-level GAT message passing	Hierarchical graph: Intra-cluster local graphs to Inter-cluster global super-nodes	GraphCast 10-day forecasts (T, u, v, Z at multiple pressure levels) from ERA5	Nationwide air quality networks; multi-day recursive autoregressive rollouts ¹⁹
Hu et al. (2023, <i>Env. Int.</i>) [cite: 17]	Source-Receptor Hybrid GNN-LSTM	Dynamic spatial attention tracking upwind source contribution across downwind nodes	Receptor-centric network augmented with 128 regional upwind sites	Prevailing wind trajectories, regional emission corridors	72-hour real-time forecasting; source attribution growing from 0% (3h) to 38% (72h) ¹⁷
MGAtt-LSTM (Bo et al., 2024, <i>Env. Modell. Softw.</i>) [cite: 21, 38]	Multi-Scale Multi-Graph Attention Network	Decoupled multi-head attention across geographic, meteorological, and regional matrices	Tri-graph architecture: Geographic, Meteorological, and Regional	Temperature, pressure, wind velocity, regional connectivity	Multi-site urban basins; evaluated across advection vs stagnation regimes ²¹
DeepRSM (Xing et al., 2020, <i>ES&T</i>) [cite: 24, 39]	Deep Non-Linear Response Surface Surrogate	Response surface derivatives: Contribution mapping	Gridded regional chemical transport model receptor	Synoptic meteorology, regional boundary layer dynamics,	WRF-CMAQ perturbation benchmarks;

		emissions to concentrations	cells	chemical regimes	non-linear ozone and PM2.5 chemistry ($R^2 >$) ²⁴
Lakshmi et al. (2025, <i>Sci. Rep.</i>) [cite: 9]	Winter Episode Transfer Attention Network	Transfer learning pre-trained on multi-year data; fine-tuned on extreme winter peaks	Multi-head attention across temporal lag windows	NASA FIRMS Fire Radiative Power (FRP), wind speed/direction, RH, inversion	Decadal Delhi winter dataset (2012–2022); 10-fold cross-validation; head interpretability ⁹

Architectural Synthesis and System Integration

Synthesizing the core methodologies from these 15 foundational publications provides the architectural blueprint for an operational, physics-constrained, source-aware air quality forecasting system¹. The resulting system—designated the STAGE-PM framework (Source, Transport, Atmospheric stability, Dynamic Graph, Explainable operator)—replaces unconstrained spatio-temporal convolutions with a physically grounded transport architecture¹.



- Regulates Vertical Trapping vs Horizontal Venting



[Multi-Scale Dynamic Graph: GL + GR + GA]

- Intra-Urban Local Clusters (GL)
- Transboundary Regional Corridors (GR)
- Synoptic Atmospheric State (GA)



[Physics-Constrained Neural Operator Backbone]

- Decoupled Advection, Diffusion & Reaction Operators
- Boundary-Aware Open-System Dynamics (BA-DAE)



[Multi-Task Output Head]

- Continuous-Time Concentration Rollouts (+1h, +6h, +24h, +72h)
- Real-Time Quantitative Source Attribution Vector
- Calibrated Predictive Uncertainty Intervals

Transitioning from Binary Heuristics to Continuous Stability

Formulations

A critical limitation in earlier dynamic graph designs was the use of arbitrary, discontinuous threshold rules—such as disconnecting graph edges whenever boundary layer height (BLH) dropped below 100 meters¹. Atmospheric physics does not exhibit a step-function discontinuity at 100 meters; mixing efficiency varies continuously with thermal stratification, shear turbulence, and solar radiation¹.

Drawing directly from the continuous formulation principles of MA-NODE and RegBoost, the framework replaces binary rules with a continuous Atmospheric Stability Index $S_t \in [0, 1]$ ¹:

$$S_t = \sigma \left(\mathbf{w}^T [\text{BLH}_t, \Delta T_{\text{inversion},t}, \text{RH}_t, \text{WS}_t, \text{SR}_t]^T + b \right)$$

where BLH denotes boundary layer height, $\Delta T_{\text{inversion}}$ measures vertical temperature inversion strength between the surface and 850 hPa, RH is relative humidity, WS is surface wind speed, and SR is downward solar radiation¹. An index value approaching 0 indicates intense convective mixing and strong vertical ventilation, whereas a value approaching 1 represents extreme nocturnal inversion, calm winds, and surface-layer trapping¹.

The dynamic graph adjacency matrix A_{ij}^t between source node i and receptor node j is

then continuously modulated by this stability function¹:

$$A_{ij}^t = A_{ij}^{\text{wind},t} \cdot \mathcal{G}(S_t) \cdot \tau_{ij}^t$$

When $S_t \rightarrow 1$, the continuous modulation function $\mathcal{G}(S_t)$ attenuates long-distance horizontal message passing and amplifies local self-loops, directly simulating the physical trapping of urban emissions beneath a shallow thermal inversion lid¹. Conversely, when $S_t \rightarrow 0$, regional advective edges expand, enabling long-range regional transport across the graph¹.

Embedding Physical Transport Travel Time and Probabilistic Corridors

Conventional spatial message passing assumes instantaneous information transfer between connected nodes, ignoring fluid travel times¹. To model the movement of agricultural smoke from Punjab and Haryana into Delhi, the framework integrates the dynamic transport lag principles demonstrated by Hu et al. and Lakshmi et al.¹.

The dynamic transport travel delay $\tau_{s,i}(t)$ between regional source s and urban receptor i is calculated from path-integrated wind fields¹:

$$\tau_{s,i}(t) = \frac{d_{s,i}}{U_{\text{eff}}(t)}$$

where $d_{s,i}$ is the effective downwind path distance and $U_{\text{eff}}(t)$ is the effective wind transport velocity along the transport corridor¹. Instead of correlating current receptor concentrations with instantaneous source emissions, the message-passing operator aggregates time-lagged source states:

$$E_s(t - \tau_{s,i}(t)) \longrightarrow C_i(t)$$

This mechanism allows the neural network to capture the physical transit of plumes: elevated fire activity in Punjab at time $t - \tau$ systematically elevates downstream Delhi monitoring stations at time t ¹.

Furthermore, deterministic edges connecting regional sources to urban stations are replaced with Probabilistic Transport Corridors, adopting the directional projection principles of Zhao et al. and TransNet¹:

$$P(i, s, t) = \text{Softmax} \left(\frac{\mathbf{u}_s(t) \cdot \mathbf{e}_{si}}{\|\mathbf{u}_s(t)\|} \cdot \exp \left(-\frac{d_{si}^2}{2\sigma_d^2} \right) \cdot \mathcal{G}(S_t) \right)$$

where $\mathbf{e}_{si}$ represents the unit direction vector pointing from source s to receptor i . This formulation assigns each source-receptor pair a continuous, wind- and stability-dependent transport probability, reflecting the turbulent expansion of plumes as they travel across the airshed¹.

Tri-Layer Multi-Scale Graph Topology

Following the hierarchical and multi-graph designs of GraphCast-GNN and MGAtt-LSTM, the spatial architecture is formalized into three distinct graph layers¹:

$$G(t) = G_L(t) \oplus G_R(t) \oplus G_A(t)$$

The local intra-urban graph $G_L(t)$ interconnects individual CAAQMS monitoring stations across Delhi, using static road-network densities and local aerodynamic roughness lengths to model street-level vehicle accumulation, micro-scale eddy dispersion, and local industrial emissions¹.

The regional transport graph $G_R(t)$ connects external source nodes (Punjab, Haryana, Rajasthan, and western Uttar Pradesh) to urban receptor clusters, tracking agricultural biomass burning, desert dust advection, and long-range industrial transport¹.

The atmospheric state graph $G_A(t)$ operates as a continuous latent hyper-graph that encodes boundary layer height, vertical temperature gradients, synoptic pressure systems, and relative humidity, acting as a global conditioning filter that modulates edge weights across both the local and regional graphs¹.

Physics-Constrained Regularization Loss

To prevent the neural operator from learning spurious correlations or violating fluid continuity, the optimization objective couples prediction accuracy with physical conservation constraints, synthesizing the loss formulations of TransNet, CTENet, Air-DualODE, and PI-GNN¹:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{forecast}} + \lambda_1 \mathcal{L}_{\text{transport}} + \lambda_2 \mathcal{L}_{\text{mass}} + \lambda_3 \mathcal{L}_{\text{stability}}$$

The empirical loss $\mathcal{L}_{\text{forecast}}$ evaluates multi-step point and quantile prediction errors across

monitoring stations¹.

The transport loss $\mathcal{L}_{\text{transport}}$ penalizes spatial concentration gradients that contradict the wind velocity vector field:

$$\mathcal{L}_{\text{transport}} = \left\| \frac{\partial \hat{C}}{\partial t} + \mathbf{u} \cdot \nabla \hat{C} - K \nabla^2 \hat{C} - S + D \right\|_2^2$$

The mass conservation loss $\mathcal{L}_{\text{mass}}$ enforces continuity across the open boundaries of the airshed, penalizing unmodeled mass generation or destruction:

$$\mathcal{L}_{\text{mass}} = \sum_t \left| \int_{\Omega} \frac{\partial \hat{C}}{\partial t} d\Omega + \oint_{\partial\Omega} (\hat{C} \mathbf{u}) \cdot \mathbf{n} d\Gamma - \sum \hat{S}_{\text{emissions}} + \sum \hat{D}_{\text{sinks}} \right|$$

where Ω represents the airshed spatial domain, $\partial\Omega$ denotes its perimeter, and $\mathbf{n}$ is the outward normal unit vector¹⁰.

Finally, the stability loss $\mathcal{L}_{\text{stability}}$ penalizes vertical transport when $S_t \rightarrow 1$, ensuring that the model maintains localized surface accumulation during severe thermal inversions¹.

Explainable Source Attribution and Event-Aware Regime Switching

To provide actionable insights for environmental policymakers, the model augments scalar concentration forecasts with an interpretable Source Contribution Head, adapting the non-linear response formulations of DeepRSM¹. Alongside future particulate concentrations, the output layer generates a normalized source attribution vector:

$$\mathbf{y}_i(t + \Delta t) = \left[\hat{C}_i(t + \Delta t), \hat{\pi}_{\text{traffic}}, \hat{\pi}_{\text{industry}}, \hat{\pi}_{\text{biomass}}, \hat{\pi}_{\text{dust}}, \hat{\pi}_{\text{background}} \right]$$

subject to the physical unit-sum constraint $\sum_k \hat{\pi}_k = 1.0$ ¹. This mechanism explains *why* pollution is elevated, quantifying whether an impending hazard is driven by local vehicular emissions, regional stubble burning, or transboundary desert dust¹.

Simultaneously, the architecture incorporates an Event-Aware Regime Detector inspired by the multi-timescale latent dynamics of MA-NODE and FFTGNet¹. A classification network evaluates

synoptic atmospheric states and regional emission proxies to predict the active pollution regime:

$$P(\text{Regime}_t) = \text{Softmax}(\mathbf{W}_{\text{regime}} \mathbf{z}_{\text{synoptic}}(t))$$

classifying conditions into Normal Urban, Inversion Trapping, Biomass Burning Plume, or Dust Advection¹. The predicted regime dynamically adjusts the message-passing weights within the neural operator: under Inversion Trapping, vertical dispersion is restricted and local edge weights dominate; under Biomass Burning, upwind transport corridors from the northwest are prioritized; and under Dust Advection, coarse particulate channels and western boundary inputs are amplified¹.

Strategic Integration of Literature into Framework Components

The table below maps each component of the proposed framework directly to the foundational papers that supply its mathematical, structural, and empirical justification:

Proposed System Component	Supporting Literature	Methodological Contribution & Applied Mechanism
Continuous Atmospheric Stability Modulation	<i>RegBoost</i> ²² , <i>MA-NODE</i> ³⁴ , <i>Kim et al.</i> ¹⁶	Derives a continuous stability scalar $S_t =$ that scales edge weights smoothly, eliminating crude binary cutoffs ¹ .
Dynamic Transport Travel Lag ($\tau_{s,i}(t)$)	<i>Hu et al.</i> ¹⁷ , <i>Lakshmi & Krishnamoorthy</i> ⁹	Formulates plume travel delays as $\tau = d/U_{\text{eff}}$, linking upstream agricultural fires in Punjab directly to delayed urban peaks in Delhi ¹ .
Probabilistic Transport Corridors	<i>Zhao et al.</i> ² , <i>UrbanGraph</i> ¹⁵ , <i>TransNet</i> ⁴	Builds dynamic directed graphs where edge connectivity reflects wind vector projections and spatial plume dispersion rather than static Euclidean

		distance ¹ .
Tri-Layer Spatial Hierarchy (G_L, G_R, G_A)	<i>GraphCast-GNN</i> ¹⁹ , <i>MGAAtt-LSTM</i> ²¹ , <i>UrbanGraph</i> ¹⁵	Decouples local urban street-level diffusion from transboundary transport corridors and synoptic atmospheric stability states ¹ .
Coupled ADR Neural Operator Backbone	<i>TransNet</i> ⁴ , <i>AirPhyNet</i> ¹² , <i>CTENet</i> ¹³	Replaces standard graph attention with message passing that embeds the continuous advection-diffusion-reaction partial differential equations ¹ .
Open-Boundary Mass Conservation Regularizer	<i>Air-DualODE</i> ¹⁰ , <i>PI-GNN</i> ¹⁸ , <i>STeP-Diff</i> ²³	Enforces mass conservation across airshed boundaries through differential loss functions, penalizing unphysical particulate creation or dissipation ¹⁰ .
Real-Time Source Attribution Head	<i>DeepRSM</i> ²⁴ , <i>Hu et al.</i> ¹⁷	Uses response-surface derivative formulations to predict continuous source contribution percentages alongside total particulate mass ¹ .
Event-Aware Regime Switching	<i>MA-NODE</i> ³⁴ , <i>FFTGNet</i> ²⁰ , <i>Lakshmi et al.</i> ⁹	Decomposes multi-scale temporal dynamics to classify synoptic pollution regimes and dynamically route network operations during extreme episodes ¹ .
Satellite Fire Radiative Power Proxy	<i>Lakshmi et al.</i> ⁹ , <i>RegBoost</i> ²²	Converts satellite thermal detections into an integrated Fire Emission

		Proxy ($FEP = \sum FRP \times Conf$) instead of using raw fire counts ¹ .
Calibrated Predictive Uncertainty	MA-NODE ³⁴ , STeP-Diff ²³	Produces continuous predictive distributions and calibrated 95% uncertainty intervals, enabling probabilistic threshold exceedance forecasting ¹ .

Critical Synthesis and Future Directions

A comparative analysis of these 15 foundational studies reveals why conventional, unconstrained machine learning architectures consistently fail during extreme air pollution episodes, while clarifying how physics-constrained, source-aware systems resolve these failure modes¹.

Standard spatio-temporal graph neural networks rely on static geographical distance metrics or isotropic attention mechanisms, implicitly assuming that spatial dependencies are stationary and symmetric¹. In real atmospheric systems, these assumptions are invalid⁴. Under high-speed advective winds, an upwind emission source located 100 kilometers away exerts immediate downstream influence, whereas an adjacent monitoring station located 2 kilometers cross-wind remains completely uncoupled².

Conversely, during strong winter thermal inversions, boundary layer heights collapse below 150 meters and horizontal wind speeds drop below 1 m/s, causing horizontal transport corridors to close entirely²². Under these conditions, the regional airshed fragments into localized accumulation pockets dominated by ground emissions and surface-layer chemical reactions¹. Models that rely on static graphs or unconstrained attention overfit to moderate seasonal averages⁴. As a result, they systematically underpredict hazardous episodic peaks and miscalculate the timing of meteorological clearances⁴.

The literature demonstrates that addressing these challenges requires decomposing monolithic neural layers into decoupled physical operators⁴. TransNet, AirPhyNet, and UrbanGraph confirm that message-passing architectures must independently parameterize advective flux, turbulent diffusion, and reactive transformation⁴. Enforcing this operator decoupling stabilizes training dynamics, eliminates spatial oversmoothing, and ensures physical plausibility⁴.

Furthermore, Air-DualODE, CTENet, and PI-GNN show that urban airsheds cannot be modeled as closed systems¹⁰. Incorporating boundary-aware flux formulations and mass conservation regularizers restricts the neural hypothesis space to solutions that obey physical conservation laws, preventing spurious mass creation and ensuring reliable generalization to unobserved

monitoring locations¹⁰.

Finally, the work of Xing et al. (DeepRSM), Hu et al., and Lakshmi et al. emphasizes that air quality forecasting must transition from isolated station correlation to causal, source-to-receptor attribution¹. Monitoring stations are environmental receptors, not primary emission sources¹. Integrating regional source nodes—parameterized through satellite-derived

Fire Radiative Power proxies, dynamic transport travel delays ($\tau = d/U_{\text{eff}}$), and continuous stability indices—allows models to capture the physical lifecycle of airborne pollutants: emission, transport, dispersion, accumulation, and chemical aging¹.

By synthesizing these 15 foundational methodologies, environmental machine learning moves beyond generic black-box curve fitting toward scientifically grounded, explainable, and operationally reliable physical AI for atmospheric systems¹.

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